

Digital Logic Design

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Digital Logic Design

The **digital logic design** is a system in electrical and computer engineering that **uses simple numerical values to produce input and output operations**. As a **digital design engineer**, you may assist in **developing cell phones, computers, and related personal electronic devices**. In the modern world of electronics, the term digital is usually associated with a computer. This is due to the fact that the term digital is derived from the way the computers perform operations, by counting digits. Nowadays digital concepts are usually applied to various problems which are solved by analog methods. In modern times digital systems find applications related to industrial and consumer products which maintain and develop various ways which develop our country and helps in developing science ideas in terms of research. As computer science is upgraded in every way it helps to develop the multinational companies in the form of bonus machines.

Numerical Representations

In science, technology, business, and, in fact, most other fields of our real-life, we are constantly dealing with ***quantities***. Quantities are measured, monitored, recorded, manipulated arithmetically, observed, or in some other way utilized in most physical systems. It is **important when dealing with various quantities** that we be able to represent their values efficiently and accurately. There are basically **two ways of representing the numerical value of quantities**:

- **Analog** and
- **Digital**

Analog Representations

In analog representation a quantity is represented by a **continuously variable, proportional indicator**. An example is an **automobile speedometer** from the classic muscle cars of the 1960s and 1970s. The deflection of the needle is proportional to the speed of the car and follows any changes that occur as the vehicle speeds up or slows down. **Thermometers** before the digital revolution used analog representation to measure temperature, and many are still in use today.

Analog quantities **have an important characteristic**, no matter how they are represented: ***they can vary over a continuous range of values***. The automobile speed can have *any* value between zero and, say, 100 mph. Similarly, the microphone output might have any value within a range of **zero to 10 mV (e.g., 1 mV, 2.3724 mV, 9.9999 mV)**.

Digital Representations

In digital representation the quantities are represented not by **continuously variable indicators but by symbols called *digits***. As an example, consider the **digital clock**, which provides the time of day in the form of **decimal digits (i.e., *discrete steps*)** that represent hours and minutes (and sometimes seconds). As we know, the **time-of-day changes continuously**, but the **digital clock reading does not change continuously**; rather, it **changes in steps** of one per minute (or per second).

The **major difference** between analog and digital quantities, then, can be simply stated as follows:

- **Analog = Continuous**
- **Digital = Discrete (step by step)**

Because of the discrete nature of **digital representations**, there is **no ambiguity** when reading the value of a digital quantity. Whereas the value of an **analog quantity** is often **leads to ambiguity**. In practice, when we take a measurement of an analog quantity, we always **“round” to a convenient level of precision**. In other words, **we digitize the quantity**. For example, when you take your **temperature with a mercury (analog) thermometer**, the mercury column is usually between two graduation lines, but you would pick the nearest line and assign it a number of, say, 98.6°F.

Example 1: Which of the following involve analog quantities and which involve digital quantities?

- (a) Ten-position switch
- (b) Current flowing from an electrical switch
- (c) Temperature of a room
- (d) Sand grains on the beach
- (e) Automobile fuel gauge

Solution:

- (a) Digital
- (b) Analog
- (c) Analog
- (d) Digital, since the number of grains can be only certain discrete (integer) values and not every possible value over a continuous range

(e) Analog, if needle type; digital, if numerical readout or bar graph display

Digital and Analog Systems

A **digital system** is a **combination of devices** designed to **manipulate logical information or physical quantities** that are represented in **digital form**; that is, the quantities can take on only **discrete values**. These **devices** are most often **electronic**, but they can also be **mechanical, magnetic, or pneumatic**. Some of the **familiar digital systems include: digital computers and calculators, digital audio and video equipment, and the telephone system-the world's largest digital system.**

An **analog system** contains devices that **manipulate physical quantities** that are represented in **analog form**. In an analog system, the **quantities can vary over a continuous range of values**. For example, the amplitude of the output signal to the speaker in a radio receiver can have any value between zero and its maximum limit. Other **common analog systems are: audio amplifiers, magnetic tape recording and playback equipment, and a simple light dimmer switch.**

Advantages of Digital Techniques

An increasing majority of applications in electronics, as well as in most other technologies, use digital techniques to perform operations that were once performed using analog methods. The **chief reasons for the shift to digital technology** are:

- 1. Digital systems are generally easier to design.** The circuits used in digital systems are *switching circuits*, where *exact* values of voltage or current are not important, only the **range (HIGH or LOW)** in which they fall.
- 2. Information storage is easy.** This is accomplished by special devices and circuits that can latch onto digital information and hold it for as long as necessary, and mass storage techniques that **can store billions of bits of information in a relatively small physical space**. Analog storage capabilities are, by contrast, extremely limited.
- 3. Accuracy and precision are easier to maintain throughout the system.** Once a signal is digitized, the **information it contains does not deteriorate** as it is processed. In **analog systems, the voltage and current signals tend to be distorted** by the effects of temperature, humidity, and component tolerance variations in the circuits that process the signal.
- 4. Operation can be programmed.** It is fairly easy to design digital systems whose operation is controlled by a set of stored instructions called a *program*.

Analog systems can also be *programmed*, but the variety and the complexity of the available operations are severely limited.

5. Digital circuits are less affected by noise. Spurious fluctuations in voltage (noise) are not as critical in digital systems because the exact value of a voltage is not important, as long as the **noise is not large enough to prevent us from distinguishing a HIGH from a LOW.**

6. More digital circuitry can be fabricated on IC chips. It is true that analog circuitry has also benefited from the tremendous development of IC technology, but its relative complexity and its use of devices that cannot be economically integrated (high-value capacitors, precision resistors, inductors, transformers) have prevented analog systems from achieving the same high degree of integration.

Limitations of Digital Techniques

There are really very few drawbacks when using digital techniques. The **two biggest problems** are:

- i. **The real world is analog.**
- ii. **Processing digitized signals takes time.**

Most **physical quantities are analog in nature**, and these quantities are often the inputs and outputs that are being monitored, operated on, and controlled by a system. Some examples are **temperature, pressure, position, velocity, liquid level, flow rate**, and so on. We are in the **habit of expressing these quantities digitally**, such as when we say that the temperature is (when we want to be more precise), but we are really **making a digital approximation to an inherently analog quantity**. Therefore, **to take advantage of digital techniques when dealing with analog inputs and outputs, four steps must be followed:**

1. Convert the physical variable to an electrical signal (analog).
2. Convert the electrical (analog) signal into digital form.
3. Process (operate on) the digital information.
4. Convert the digital outputs back to real-world analog form.

The Present and Future is Digital

The advances in digital technology over the past three decades have been nothing short of phenomenal, and there is every reason to believe that more is coming. Think of the everyday items that have changed from analog format to digital in your lifetime.

- An indoor/outdoor wireless **digital thermometer** can be purchased for less than or around TK 100.
- **Automobile Cars** have gone from having very few electronic controls to being predominantly digitally controlled vehicles. Maybe your automobile is equipped with systems which turns your dashboard into a hub for **wireless communication, information, and navigation**. You may already be using **voice commands** to send or retrieve e-mail, call for a traffic report, check on the car's maintenance needs, or just switch radio stations or CDs—all without taking your hands off the wheel or your eyes off the road. Cars can **report their exact location** in case of emergency or mechanical breakdown.
- Digital **audio** has moved us to the compact disk and MP3 player. Digital **video** brought the DVD.
- **Digital home video** and still **cameras; digital recording** with systems like TiVo; **digital cellular phones; and digital imaging in X-Ray, magnetic resonance imaging (MRI), and ultrasound** systems in hospitals are just a few of the applications that have been taken over by the digital revolution.
- In the coming years wireless communication will continue to expand coverage to provide connectivity wherever you are. **Telephones** will be able to **receive, sort, and maybe respond to incoming calls** like a well-trained secretary.
- The **digital television** revolution will provide not only higher definition of the picture, but also much more **flexibility in programming**. You will be able to **select the programs** that you want to view and **load them into your television's memory**, allowing you to pause or replay scenes at your convenience, very much like viewing a DVD today.
- As **virtual reality** continues to improve, you will be able to interact with the subject matter you are studying. This may not sound exciting when studying electronics, but imagine studying history from the standpoint of being a participant, or learning proper techniques for everything from athletics to surgery through simulations based on your actual performance.

Digital technology will continue its high-speed incursion into current areas of our lives as well as **break new ground in ways we may never have considered**. The **software tools to develop complex systems are constantly being upgraded** and are available to anyone over the Web.

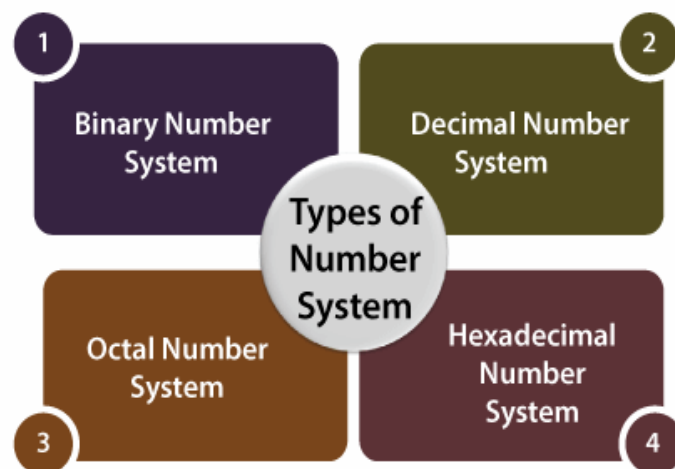
Number System

In a **digital system**, the system can **understand only the optional number system**. In these systems, **digits symbols are used to represent different values**, depending on the index from which it settled in the number system. In simple terms, **for representing the information, we use the number system in the digital system**. The **digit value** in the number system is **calculated using**:

- The **digit**
- The **index**, where the digit is present in the number.
- Finally, the **base numbers**, the total number of digits available in the number system.

Types of Number System

In the digital computer, there are various types of number systems used for representing information. The majors are shown in below figure:



1) Binary Number System

- Generally, a binary number system is **used in the digital computers**.
- In this number system, it **carries only two digits, either 0 or 1**.
- There are **two types of electronic pulses** present in a binary number system. The first one is the **absence of an electronic pulse representing '0'** and second one is the **presence of electronic pulse representing '1'**.
- **Each digit is known as a bit**. A four-bit collection (1101) is known as a **nibble**, and a collection of eight bits (11001010) is known as a **byte**.

- It is also known as the **base 2 number system**. The **location of a digit** in a binary number represents a **specific power of the base (2)** of the number system.

Examples:

$(10100)_2$, $(11011)_2$, $(11001)_2$, $(000101)_2$, $(011010)_2$.

2) Decimal Number System

- The decimal numbers are **used in our day-to-day life**.
- The decimal number system **contains ten digits from 0 to 9** (base 10). The **0 is the minimum value** of the digit, and **9 is the maximum value** of the digit.
- Here, the **successive place value or position, left to the decimal point** holds **units, tens, hundreds, thousands**, and so on.
- The **position** in the decimal number system **specifies the power of the base (10)**. For example, the decimal number **2541** consist of the digit **1 in the unit position, 4 in the tens position, 5 in the hundreds position, and 2 in the thousand positions** and the value will be written as:

$$\begin{aligned}
 &= (2 \times 1000) + (5 \times 100) + (4 \times 10) + (1 \times 1) \\
 &= (2 \times 10^3) + (5 \times 10^2) + (4 \times 10^1) + (1 \times 10^0) \\
 &= 2000 + 500 + 40 + 1 \\
 &= 2541
 \end{aligned}$$

3) Octal Number System

- The octal number system has **base 8** (means it has only **eight digits from 0 to 7**).
- With the **help of only three bits**, an **octal number is represented**. Each set of bits has a distinct value between 0 and 7.

Octal Number	Binary Number
0	000
1	001
2	010
3	011

4	100
5	101
6	110
7	111

Examples:

$(273)_8$, $(5644)_8$, $(0.5365)_8$, $(1123)_8$, $(1223)_8$.

4) Hexadecimal Number System

- The number system has a **base of 16** means there are **total 16 symbols** (0, 1, 2, 3, 4, 5, 6, 7, 8, 9, A, B, C, D, E, F) used for representing a number. It has **ten digits from 0 to 9** and **6 letters from A to F**. The single-bit representation of decimal **values 10, 11, 12, 13, 14, and 15** are represented by **A, B, C, D, E, and F**.
- Only **4 bits are required** for representing a number in a **hexadecimal number**. Each set of bits has a distinct value between 0 and 15.

Binary Number	Hexadecimal Number
0000	0
0001	1
0010	2
0011	3
0100	4
0101	5
0110	6
0111	7
1000	8
1001	9
1010	A

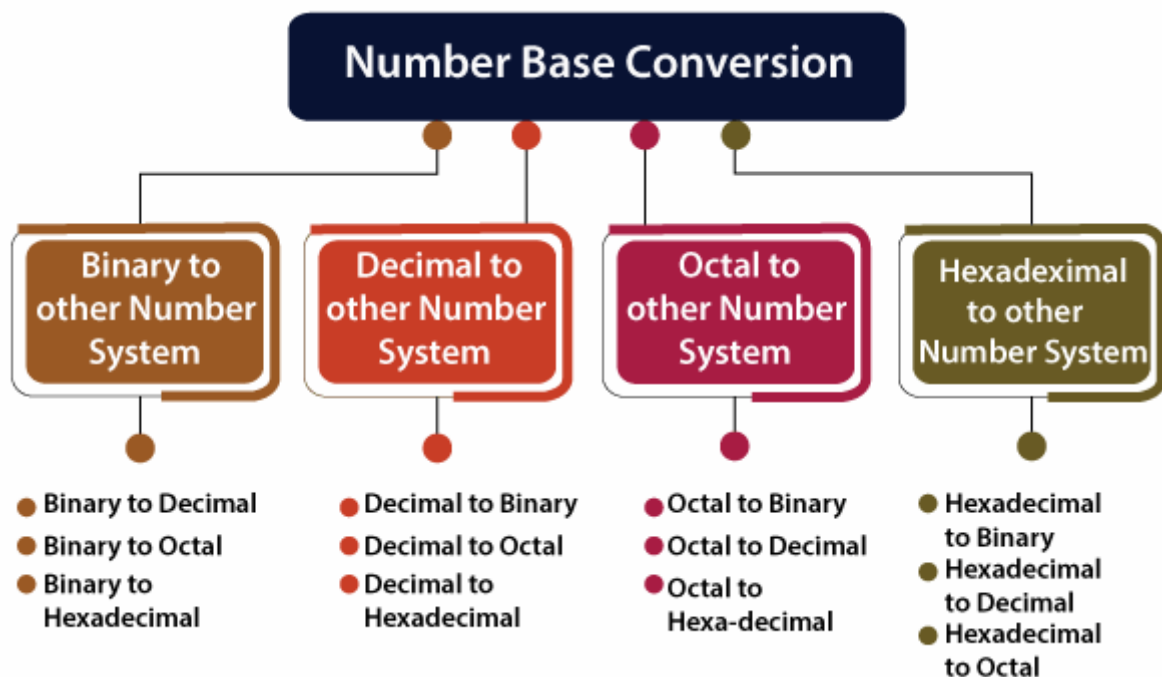
1011	B
1100	C
1101	D
1110	E
1111	F

Examples:

$(FAC2)_{16}$, $(564)_{16}$, $(0ABD5)_{16}$, $(1123)_{16}$, $(11F3)_{16}$.

Number Base Conversion

In our previous section, we learned different types of number systems such as binary, decimal, octal, and hexadecimal. In this part of the tutorial, we will **learn how we can change a number from one number system to another number system**. As, we have **four types of number systems** so **each one can be converted into the remaining three systems**. There are the following conversions possible in Number System:



Decimal to other Number Systems

The decimal number can be an integer or floating-point integer. When the decimal number is a floating-point integer, then we convert both part (integer and fractional) of the decimal number in the isolated form (individually).

Decimal to Binary Conversion

The **following two types of operations take place**, while converting decimal number into its equivalent binary number.

- **Division of integer part** and successive quotients with base 2.
- **Multiplication of fractional part** and successive fractions with base 2.

Example

Consider the **decimal number 58.25**. Here, the integer part is 58 and fractional part is 0.25.

Step 1 – Division of 58 and successive quotients with base 2.

Operation	Quotient	Remainder
58/2	29	0 (LSB)
29/2	14	1
14/2	7	0
7/2	3	1
3/2	1	1
1/2	0	1 (MSB)

$$58_{10} = 111010_2$$

Therefore, the **integer part** 58 of equivalent binary number is **111010**.

Step 2 – Multiplication of 0.25 and successive fractions with base 2.

Operation	Result	Carry
0.25 x 2	0.5	0
0.5 x 2	1.0	1
-	0.0	-

$$.25_{10} = .01_2$$

Therefore, the **fractional part** of equivalent binary number is **.01**

$$58.25_{10} = 111010.01_2$$

Therefore, the **binary equivalent** of decimal number 58.25 is 111010.01

Decimal to Octal Conversion

The following two types of operations take place, while converting decimal number into its equivalent octal number.

- **Division of integer part** and successive quotients with base 8.
- **Multiplication of fractional part** and successive fractions with base 8.

Example

Consider the **decimal number 58.25**. Here, the integer part is 58 and fractional part is 0.25.

Step 1 – Division of 58 and successive quotients with base 8.

Operation	Quotient	Remainder
58/8	7	2
7/8	0	7

$$58_{10} = 72_8$$

Therefore, the **integer part** of equivalent octal number is **72**.

Step 2 – Multiplication of 0.25 and successive fractions with base 8.

Operation	Result	Carry
0.25 x 8	2.00	2
-	0.00	-

$$.25_{10} = .2_8$$

Therefore, the **fractional part** of equivalent octal number is .2

$$58.25_{10} = 72.2_8$$

Therefore, the **octal equivalent** of decimal number 58.25 is 72.2.

Decimal to Hexa-Decimal Conversion

The following two types of operations take place, while converting decimal number into its equivalent hexa-decimal number.

- **Division of integer part** and successive quotients with base 16.
- **Multiplication of fractional part** and successive fractions with base 16.

Example

Consider the **decimal number 58.25**. Here, the integer part is 58 and decimal part is 0.25.

Step 1 – Division of 58 and successive quotients with base 16.

Operation	Quotient	Remainder
58/16	3	10=A
3/16	0	3

$$58_{10} = 3A_{16}$$

Therefore, the **integer part** of equivalent Hexa-decimal number is 3A.

Step 2 – Multiplication of 0.25 and successive fractions with base 16.

Operation	Result	Carry
0.25 x 16	4.00	4
-	0.00	-

$$.25_{10} = .4_{16}$$

Therefore, the **fractional part** of equivalent Hexa-decimal number is .4.

$$58.25_{10} = 3A.4_{16}$$

Therefore, the **Hexa-decimal equivalent** of decimal number 58.25 is 3A.4.

Binary to other Number Systems

There are three conversions possible for binary number, i.e., binary to decimal, binary to octal, and binary to hexadecimal. The conversion process of a binary number to decimal differs from the remaining others.

Binary to Decimal Conversion

The process of converting binary to decimal is quite simple. The process **starts from multiplying the bits of binary number with its corresponding positional weights**. And lastly, we **add all those products**. Let's take an example to understand how the conversion is done from binary to decimal.

Example 1: (10110.001)₂

We multiplied each bit of (10110.001)₂ with its respective positional weight, and last, we add the products of all the bits with its weight.

$$\begin{aligned}
 (10110.001)_2 &= (1 \times 2^4) + (0 \times 2^3) + (1 \times 2^2) + (1 \times 2^1) + (0 \times 2^0) + (0 \times 2^{-1}) + (0 \times 2^{-2}) + (1 \times 2^{-3}) \\
 &= (1 \times 16) + (0 \times 8) + (1 \times 4) + (1 \times 2) + (0 \times 1) + (0 \times 1/2) + (0 \times 1/4) + (1 \times 1/8) \\
 &= 16 + 0 + 4 + 2 + 0 + 0 + 0 + 0.125 \\
 &= (22.125)_{10}
 \end{aligned}$$

Binary to Octal Conversion

The base numbers of binary and octal are 2 and 8, respectively. In a binary number, the pair of three bits is equal to one octal digit. There are **only two steps to convert a binary number into an octal number** which are as follows:

- In the first step, we have to **make the pairs of three bits** on both sides of the binary point. If there will be one or two bits left in a pair of three bits pair, we add the required number of zeros on extreme sides.
- In the second step, we **write the octal digits corresponding to each pair**.

Example 1: $(111110101011.0011)_2$

1. Firstly, we make pairs of three bits on both sides of the binary point.

111 110 101 011.001 1

On the right side of the binary point, the last pair has only one bit. To make it a complete pair of three bits, we added two zeros on the extreme side.

111 110 101 011.001 100

2. Then, we wrote the octal digits, which correspond to each pair.

$$(111110101011.0011)_2 = (7653.14)_8$$

Binary to Hexadecimal Conversion

The base numbers of binary and hexadecimal are 2 and 16, respectively. In a binary number, the pair of four bits is equal to one hexadecimal digit. There are also **only two steps to convert a binary number into a hexadecimal number** which are as follows:

- In the first step, we have to **make the pairs of four bits** on both sides of the binary point. If there will be one, two, or three bits left in a pair of four bits pair, we add the required number of zeros on extreme sides.
- In the second step, we **write the hexadecimal digits corresponding to each pair**.

Example 1: $(10110101011.0011)_2$

1. Firstly, we make pairs of four bits on both sides of the binary point.

111 1010 1011.0011

On the left side of the binary point, the first pair has three bits. To make it a complete pair of four bits, add one zero on the extreme side.

0111 1010 1011.0011

2. Then, we write the hexadecimal digits, which correspond to each pair.

$$(011110101011.0011)_2 = (7AB.3)_{16}$$

Octal to other Number System

Like binary and decimal, the octal number can also be converted into other number systems. The process of converting octal to decimal differs from the remaining one.

Octal to Decimal Conversion

The process of converting octal to decimal is the same as binary to decimal. The process starts from **multiplying the digits of octal numbers with its corresponding positional weights**. And lastly, we **add all those products**. Let's take an example to understand how the conversion is done from octal to decimal.

Example 1: $(152.25)_8$

Step 1:

We multiply each digit of **152.25** with its respective positional weight, and last, we add the products of all the bits with its weight.

$$\begin{aligned}(152.25)_8 &= (1 \times 8^2) + (5 \times 8^1) + (2 \times 8^0) + (2 \times 8^{-1}) + (5 \times 8^{-2}) \\ &= 64 + 40 + 2 + (2 \times 1/8) + (5 \times 1/64) \\ &= 64 + 40 + 2 + 0.25 + 0.078125 \\ &= 106.328125\end{aligned}$$

So, the decimal number of the octal number 152.25 is **$(106.328125)_{10}$**

Octal to Binary Conversion

The process of converting octal to binary is the reverse process of binary to octal. We **write the three bits binary code of each octal number digit**.

Example 1: $(152.25)_8$

We write the three-bit binary digit for 1, 5, 2, and 5.

$$(152.25)_8 = (001101010.010101)_2$$

So, the binary number of the octal number 152.25 is **$(001101010.010101)_2$**

Octal to Hexadecimal Conversion

For converting octal to hexadecimal, there are two steps required to perform, which are as follows:

- In the first step, we will **find the binary equivalent of given octal number.**
- Next, we have to **make the pairs of four bits on both sides of the binary point.** If there will be one, two, or three bits left in a pair of four bits pair, we add the required number of zeros on extreme sides and write the hexadecimal digits corresponding to each pair.

Example 1: $(152.25)_8$

Step 1:

We write the three-bit binary digit for 1, 5, 2, and 5.

$$(152.25)_8 = (001101010.010101)_2$$

So, the binary number of the octal number 152.25 is **$(001101010.010101)_2$**

Step 2:

Now, we make pairs of four bits on both sides of the binary point.

0 0110 1010.0101 01

On the left side of the binary point, the first pair has only one digit, and on the right side, the last pair has only two-digit. To make them complete pairs of four bits, add zeros on extreme sides.

0000 0110 1010.0101 0100

Now, we write the hexadecimal digits, which correspond to each pair.

$$(0000 \quad 0110 \quad 1010.0101 \quad 0100)_2 = (6A.54)_{16}$$

Hexa-decimal to other Number System

Like binary, decimal, and octal, hexadecimal numbers can also be converted into other number systems. The process of converting hexadecimal to decimal differs from the remaining one.

Hexa-decimal to Decimal Conversion

The process of converting hexadecimal to decimal is the same as binary to decimal. The process **starts from multiplying the digits of hexadecimal numbers with its corresponding positional weights**. And lastly, we **add all those products**.

Example 1: $(152A.25)_{16}$

Step 1:

We multiply each digit of **152A.25** with its respective positional weight, and last, we add the products of all the bits with its weight.

$$\begin{aligned}
 (152A.25)_{16} &= (1 \times 16^3) + (5 \times 16^2) + (2 \times 16^1) + (A \times 16^0) + (2 \times 16^{-1}) + (5 \times 16^{-2}) \\
 &= (1 \times 4096) + (5 \times 256) + (2 \times 16) + (10 \times 1) + (2 \times 16^{-1}) + (5 \times 16^{-2}) \\
 &= 4096 + 1280 + 32 + 10 + (2 \times 1/16) + (5 \times 1/256) \\
 &= 5418 + 0.125 + 0.125 \\
 &= 5418.14453125
 \end{aligned}$$

So, the decimal number of the hexadecimal number 152A.25 is: **$(5418.14453125)_{10}$**

Hexadecimal to Binary Conversion

The process of converting hexadecimal to binary is the reverse process of binary to hexadecimal. We **write the four bits binary code of each hexadecimal number digit**.

Example 1: $(152A.25)_{16}$

We write the four-bit binary digit for 1, 5, A, 2, and 5.

$$(152A.25)_{16} = (0001\ 0101\ 0010\ 1010.0010\ 0101)_2$$

So, the binary number of the hexadecimal number **152.25** is **$(1010100101010.00100101)_2$**

Hexadecimal to Octal Conversion

For converting hexadecimal to octal, there are two steps required to perform, which are as follows:

- In the first step, we will **find the binary equivalent of the hexadecimal number**.

- Next, we have to **make the pairs of three bits on both sides of the binary point**. If there will be one or two bits left in a pair of three bits pair, we add the required number of zeros on extreme sides and write the octal digits corresponding to each pair.

Example 1: $(152A.25)_{16}$

Step 1:

We write the four-bit binary digit for 1, 5, 2, A, and 5.

$$(152A.25)_{16} = (0001\ 0101\ 0010\ 1010.0010\ 0101)_2$$

So, the binary number of hexadecimal number 152A.25 is $(0011010101010.010101)_2$

Step 2:

Then, we make pairs of three bits on both sides of the binary point.

$$001\ 010\ 100\ 101\ 010.001\ 001\ 010$$

Then, we write the octal digit, which corresponds to each pair.

$$(001010100101010.001001010)_2 = (12452.112)_8$$

So, the octal number of the hexadecimal number 152A.25 is **12452.112**

equivalent of Hexa-decimal number 65.4C is 145.23.